



SACS

Quality Control

Version 24.00

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1 INTRODUCTION

This document is issued for the information of users and the potential users of the Bentley SACS Software System programs. The program quality control topics covered in this document are Software Control Procedures, Testing Procedures, Software Support Services, Functional Organization Description and Documentation. The development of the quality control procedure has occurred over many years and is intended to meet the specifications of NATO Standards Document AQAP-13.

2 SOFTWARE CONTROL PROCEDURES

Bentley Systems Inc. develops and maintains the Bentley SACS Software suite at its office in Metairie, Louisiana. The program system, comprising over twenty-five (25) interrelated program modules, resides at Bentley SACS's on-site data processing facility which houses personal computers running Windows 10 & Windows 11 operating system.

Early on, in the development of the Bentley SACS System of programs Bentley Systems Inc. established management guidelines for the control of a continuously developing software product line. The central procedure for maintaining large-scale, multifaceted software that exists in an environment that requires continual innovations was to maintain all versions of the software in one set of source code. Bentley SACS has developed an internal program maintenance system that allows this to be accomplished in a highly automated procedure. Control procedures are described in the following sections.

2.1 Module Version Identification

Each program module in the Bentley SACS System is tracked throughout its life by a unique module version identification number. This ID number shows three principal module qualities. For example, in the ID #MM.UU.PP.BB

MM is the major release number

UU is the update number

PP is the patch number

BB is the build number.

Module version identifications are reported in the output listing files or, where applicable, under the help about features.

2.2 Implementation of Updates

All updates and modifications to Bentley SACS System modules are performed by Bentley SACS engineers and programmers at Bentley SACS's in-house data processing facility headquartered in Metairie, Louisiana.

2.3 Modules Release Protocol

Distribution of all Bentley SACS modules and major system releases are processed through the Module Release Group at Bentley SACS's Metairie office. Each Software release is monitored by this office which verifies testing group clearance, machine type and operating system compatibility, export and shipping requirements and any particular special client requirements.

The delivery of the program system is made through Bentley's software download center. Such distributions consist of a software installer containing all SACS modules all data files required by the program modules and the necessary support software. Also included will be a series of inputs and corresponding outputs for selected analysis models which are described in Appendix A. These analysis models have been designed to demonstrate the various functions of the program modules being

delivered. Successful installation of the program system is achieved when identical results are produced for these analysis models on the receiving computer system.

3 SOFTWARE TESTING PROCEDURES

Bentley SACS maintains a comprehensive in-house testing program. The testing program processes each newly developed or updated program module through four levels of test verification:

Level 1 Static tests

Level 2 System integration tests

Level 3 Dynamic tests

Level 4 Third-party tests

3.1 Static Tests

Static tests are carried out during the software module design and/or update phase by the engineer/programmer in charge. These tests verify the basic functioning of the software and its compatibility with the various language translators and computer operating systems. Program modules which are hardware and/or firmware dependent are checked for proper function at this time.

3.2 System Integration Tests

The Bentley SACS Suite of program modules have been designed to draw upon one common input database and one common output database. All newly generated or updated program modules are tested for compatibility with these two databases. Additionally, all program modules are verified for compatibility of input and output format with the Bentley SACS System as a whole.

3.3 Dynamic Tests

Program modules which have cleared the Static and System Integration tests are then made available to Bentley SACS's Engineering Consultancy group in Metairie. This group performs program module tests, in which test results are verified against results of completed design projects.

3.4 Third-Party Tests

In parallel with Dynamic testing, Bentley SACS issues pre-releases of the Bentley SACS System to a number of third-party test sites in Brazil, Norway, France and the United States for testing of new features and/or special requests needed on short notice.

4 SOFTWARE SUPPORT SERVICES

Bentley SACS provides an extensive software support network to supplement the Bentley SACS software quality control measures. Clients worldwide may communicate with Bentley SACS software representatives via telephone, telefax or e-mail seven days a week. The client will speak or communicate directly with one of Bentley SACS's engineers who maintains and/or uses the Bentley SACS System on a daily basis.

The interaction which results from this support system provides a continuous feedback of ideas, enhancements and improvements to the Bentley SACS system and thus maintains the state-of-the-art nature and high quality in the software product line.

5 FUNCTIONAL ORGANIZATION DESCRIPTION

Bentley SACS software development and maintenance are performed in five different phases. All phases require at least vice-presidential approval to initiate and complete. The five phases are listed in the following sections.

5.1 Program Development

This includes conceptual development as well as incorporating corrections, enhancements and modifications. This staff includes engineer/programmers and applied research engineers.

5.2 System Integration

All modifications and newly developed software are tested for conformity in internal architecture and compatibility with existing software. All new software is given initial testing to verify that the programs are performing according to the Program Development guidelines. Also, any documentation changes are noted and submitted to the Documentation Group. The programs are then submitted for Beta testing.

5.3 Beta Testing

Testing is continued by in-house engineers. This testing is performed on existing project data and compared to previous results from those projects. Also, outside third-party companies participate in Beta testing by accepting pre-releases of the program to be used on actual projects. All reports are submitted to the System Integration Group for final evaluation.

5.4 Final Testing and Release

Final testing is performed on all programs on each computer type currently supported by Bentley SACS. Standard Master Test Data is created on Bentley SACS's quality assurance server for all programs. These Master Tests are then performed on all other hardware releases and the program outputs are compared to the Master Outputs from the quality assurance server. All program releases are done in machine readable form (binary or absolute) and each release contains a complete set of installation data and output prints, described in Appendix A, to be used at each installation for quality assurance. Since it is impossible to test the programs on all hardware operating systems and compiler libraries, the final testing procedure is to be performed by the client by comparing the supplied quality test data with the test data created by the client.

5.5 Documentation

Documentation changes are normally made to coincide with new releases of the software reflecting the input from the System Integration. Documents are related to the current release of each program. Each document is identified by the date of its release as shown on the title page of each document. Documents are not updated on a page basis but are released as complete documents.

6 APPENDIX A

6.1 Quality Assurance Release Information

The following is a list of installation data and output prints provided for quality assurance testing. The final quality assurance testing is to be performed by the client by comparing the supplied quality test data with the test data created by the client.

Demo 1. Static Topside	
sacinp.demo01	SACS IV data
saclst.demo01.std	Output listing of SACS IV
rundemo01.runx	SACS run file

Demo 2. Static Offshore	
seainp.static.demo02	Seastate static load data
sacinp.demo02	SACS IV static data
jcninp.demo02	Joint Can data for static analysis
seainp.ldf.demo02	Seastate large deflection load data
pilinp.demo02	Pile input data
saclst.demo02.std	Output listing of Seastate, SACS IV and Joint Can run
ldflst.demo02.std	Output listing of large deflection analysis run
pillst.demo02.std	Output listing of single pile analysis
demo02.runx	SACS multi-file run file (runs entire demo)
static.demo02.runx	Run file for static analysis
ldf.demo02.runx	Run file for large deflection analysis
pile.demo02.runx	Run file for single-pile analysis

Demo 3. Basic Non-Linear Add-On	
sacinp.gap.demo03	SACS IV input data for Gap analysis
gapinp.demo03	Gap input data
sacinp.psi.demo03	SACS IV input data for PSI analysis
psiinp.demo03	Pile/Soil Interaction input data
gaplst.demo03.std	Output listing of nonlinear Gap analysis
psilst.demo03.std	Output listing of PSI analysis
demo03.runx	SACS multi-file run file (runs entire demo 3)
gap.demo03.runx	Run file for Gap analysis
psi.demo03.runx	Run file for PSI analysis

Demo 4a. Basic Dynamic Add-On (w/Dynamic Response)	
sacinp.demo04a	Seastate and SACS IV data
dyninp.demo04a	Dynpac modal analysis data
dyrinp.demo04a	Dynamic response (earthquake) data
saclst.demo04a.std	Output listing for static analysis run
dynlst.demo04a.std	Output listing for modal analysis run
eqklst.demo04a.std	Output listing for earthquake analysis
demo04a.runx	SACS multi-file run file (runs entire demo 4a)
static.demo04a.runx	Static analysis run file
dynpac.demo04a.runx	Modal analysis run file
earthquake.demo04a.runx	Earthquake analysis run file

Demo 4b. Basic Dynamic Add-On (w/Wave Response)	
sacinp.demo04b	SACS IV data
seainp.mode.demo04b	Seastate modal analysis load data
seainp.wave.demo04b	Seastate wave response load data
dyninp.demo04b	Modal analysis input data
wvrinp.demo04b	Wave response input data
dynlst.demo04b.std	Output listing for modal analysis run
wvrlst.demo04b.std	Output listing for wave response run
demo04b.runx	SACS multi-file run file (runs entire demo 4b)
dynpac.demo04b.runx	Modal analysis run file
wave response.demo04b.runx	Wave response run file

Demo 5a. Basic Dynamic Fatigue (w/Dynamic Response)	
sacinp.demo05a	SACS IV data
dyninp.demo05a	Dynpac modal analysis data
dyrinp.demo05a	Dynamic response (earthquake) data
ftginp.demo05a	Fatigue data
saclst.demo05a.std	Output listing for static analysis run
dynlst.demo05a.std	Output listing for modal analysis run
eqklst.demo05a.std	Output listing for earthquake analysis
ftglst.demo05a.std	Output listing for fatigue analysis
demo05a.run	SACS multi-file run file (runs entire demo 5a)
static.demo05a.runx	Static analysis run file
dynpac.demo05a.runx	Modal analysis run file
earthquake.demo05a.runx	Earthquake analysis run file
fatigue.demo05a.runx	Fatigue analysis run file

Demo 5b. Basic Dynamic Fatigue (w/Wave Response)	
sacinp.demo05b	SACS IV data
seainp.mode.demo05b	Seastate modal analysis load data
seainp.wave.demo05b	Seastate wave response load data
dyninp.demo05b	Modal analysis input data
wvrinp.demo05b	Wave response input data
dynlst.demo05b.std	Output listing for modal analysis run
wvrlst.demo05b.std	Output listing for wave response run
ftglst.demo05b.std	Output listing for fatigue analysis run
demo05b.runx	SACS multi-file run file (runs entire demo)
dynpac.demo05b.runx	Modal analysis run file
wave response.demo05b.runx	Wave response run file
fatigue.demo05b.runx	Fatigue analysis run file

Demo 6. Advanced Dynamic Fatigue	
sacinp.demo06	SACS IV data
seainp.mode.demo06	Seastate modal analysis load data
seainp.fatigue.demo06	Seastate fatigue analysis load data
dyninp.demo06	Modal analysis input data
dyrinp.demo06	Earthquake analysis input data
wvrinp.demo06	Wave response input data
ftginp.demo06	Fatigue analysis input data
saclst.demo06.std	Output listing for static analysis run
dynlst.demo06.std	Output listing for modal analysis run
eqklst.demo06.std	Output listing for earthquake analysis

wvrlst.demo06.std	Output listing for wave response run
ftglst.demo06.std	Output listing for fatigue analysis run
demo06.runx	SACS multi-file run file (runs entire demo 6)
dynpac.demo06.runx	Modal analysis run file
earthquake.demo06.runx	Earthquake analysis run file
wave response.demo06.runx	Wave response run file
fatigue.demo06.runx	Fatigue analysis run file

Demo 7. Plastic Non-Linear Add-On	
sacinp.demo07	SACS IV data
sacinp.gap.demo07	SACS IV data for nonlinear gap analysis
gapinp.demo07	Gap analysis input data
psiinp.demo07	Pile/Soil Interaction input data
clpinp.demo07	Nonlinear collapse analysis input data
saclst.demo07.std	Output listing of static analysis run
gaplst.demo07.std	Output listing of Gap analysis run
psilst.demo07.std	Output listing of PSI analysis
clplst.demo07.std	Output listing of nonlinear collapse analysis
demo07.runx	SACS multi-file run file (runs entire demo 7)
gap.demo07.runx	Gap analysis run file
psi.demo07.runx	Run file for PSI analysis
collapse.demo07.runx	Nonlinear collapse analysis run file

Demo 8. Advanced Dynamic Add-On	
sacinp.demo08	SACS IV data
seainp.demo08	Seastate input data
dyninp.demo08	Modal analysis input data
dyrinp.demo08	Earthquake analysis input data
wvrinp.demo08	Wave response input data
saclst.demo08.std	Output listing for static analysis run
dynlst.demo08.std	Output listing for modal analysis run
eqklst.demo08.std	Output listing for earthquake analysis
wvrlst.demo08.std	Output listing for wave response run
demo08.runx	SACS multi-file run file (runs entire demo8)
static.demo08.runx	Static analysis run file
dynpac.demo08.runx	Modal analysis run file
earthquake.demo08.runx	Earthquake analysis run file
wave response.demo08.runx	Wave response run file

Demo 9. Basic Marine Add-On	
sacinp.demo09	SACS IV data
gapinp.demo09	Gap analysis input data
towinp.demo09	Tow analysis input data
gaplst.demo09.std	Output listing for Gap analysis run
towlst.demo09.std	Output listing for Tow analysis run
demo09.runx	SACS multi-file run file (runs entire demo 9)
gap.demo09.runx	Gap analysis run file
tow.demo09.runx	Tow analysis run file

Demo 10. Marine Installation Add-On	
sacinp.demo10	SACS IV data
fltinp.demo10	Flotation/upending input data
lnhinp.demo10	Launch analysis input data
fltlst.demo10.std	Output listing for flotation/upending run
lnhlst.demo10.std	Output listing for Launch analysis run
demo10.runx	SACS multi-file run file (runs entire demo 10)
flotation.demo10.runx	Flotation/upending run file
launch.demo10.runx	Launch analysis run file

Demo 11. Advanced Marine Transport Add-On	
sacinp.demo11	SACS IV data
fltinp.demo11	Flotation/upending input data
lnhinp.demo11	Launch analysis input data
gapinp.demo11	Gap analysis input data
towinp.demo11	Tow analysis input data
fltlst.demo11.std	Output listing for flotation upending run
lnhlst.demo11.std	Output listing for Launch analysis run
gaplst.demo11.std	Output listing for Gap analysis run
towlst.demo11.std	Output listing for Tow analysis run
demo11.runx	SACS multi-file run file (runs entire demo 11)
flotation.demo11.runx	Flotation/upending run file
launch.demo11.runx	Launch analysis run file
gap.demo11.runx	Gap analysis run file
tow.demo11.runx	Tow analysis run file

Demo 12. Miscellaneous	
sacinp.mto.demo12	SACS IV data for Material Take-Off analysis
mtoinp.demo12	Material Take-Off data
mtolst.demo12.std	Output listing for material take-off
demo12.runx	SACS multi-file run file (runs entire demo 12)
mto.demo12.runx	Material Take-Off run file